NAGDEV PRODUCTS

**OFFICIAL TECHNICAL DATA SHEET**

**Document Ref:** NPPL-TDS-17
**Brand:** NDEVPOL™ BIO
**Origin:** North Gujarat Castor Belt, India

# Bio-Polyurethane Polyols (Castor-Oil-Based Polyols)

**UNDER DEVELOPMENT · PROPRIETARY GRADE: NDEVPOL™ BIO**

| | | | |
|---|---|---|---|
| CAS Number: | 68441-60-1 | HS Tariff Code: | 3907.29 |
| Physical Form: | Viscous Liquid | Typical Colour: | Pale Yellow to Amber |
| Feedstock / Raw Material: | Castor Oil / Castor-Oil-Derived Feedstock | Shelf Life: | 12-18 months |
| Quality System: | | | Quality systems aligned to ISO 9001 |

## ANALYTICAL PARAMETERS & SPECIFICATIONS

| SR. NO. | TEST PARAMETER | SPECIFICATION | TEST METHOD |
|---|---|---|---|
| 1 | Appearance | Clear to Slightly Hazy, Viscous Pale Yellow Liquid, free from suspended matter | Visual |
| 2 | Colour, Gardner | 4.0 Max | ASTM D1544 |
| 3 | Hydroxyl Value, mg KOH/g | 160 – 180 | ASTM D4274 |
| 4 | Acid Value, mg KOH/g | 2.0 Max | ASTM D4662 |
| 5 | Moisture Content, % | 0.20 Max | Karl Fischer / ASTM E203 |
| 6 | Viscosity at 25°C, cP | 650 – 900 | ASTM D4878 |
| 7 | Specific Gravity at 25°C | 0.950 – 0.970 | ASTM D4052 |
| 8 | Average Hydroxyl Functionality | 2.5 – 3.0 | Calculated / Standard Method |
| 9 | Iodine Value, g $I_2$/100g | 80 – 90 | AOCS Cd 1d-92 |
| 10 | Saponification Value, mg KOH/g | 175 – 190 | AOCS Cd 3-25 |
| 11 | Unsaponifiable Matter, % | 1.0 Max | AOCS Ca 6a-40 |
| 12 | Flash Point, °C | 200 Min | ASTM D92 |
| 13 | Hydroxyl Content / OH Number Consistency | Within agreed batch specification | ASTM D4274 |

## STANDARD PACKAGING & BULK LOGISTICS

200 kg HDPE / MS Drums • 1,000 kg IBC • Bulk in ISO Tank / Flexi Tank for large-volume supply • Or as recommended / specified by client

## TARGET INDUSTRIAL APPLICATIONS

Used as a renewable, bio-based polyol component in polyurethane coatings, adhesives, sealants and elastomers. A separate higher-hydroxyl-value grade should be quoted for flexible/rigid foam customers rather than using this coating-grade specification.

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*Disclaimer: The specifications above reflect typical quality values tested according to standardized AOCS / ASTM compendia. Every shipment is accompanied by a batch Certificate of Analysis (COA). HS tariff classification is indicative; final code confirmed at time of export.*

**Nagdev Products Pvt. Ltd.**
**Quality Assurance Laboratory**